

Student Number: _____



KAMBALA

Mathematics Extension 2

HSC Assessment Task 2

Half-Yearly Examination

March 2008

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Answer all questions in the writing booklets provided. **Start each question in a new booklet.**
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks – 80

- Attempt Questions 1-5.
- All questions are of equal value.

Question 1 **16 marks** **(Begin a new booklet)** **Marks**

- (a) Evaluate $|5 - 2i|$. 1
- (b) If $z = -3 - 4i$ find $\frac{1}{z}$ in the form $a + ib$. 2
- (c) Simplify $\frac{1+i^5}{1-i}$. 2
- (d) (i) Express $z = \frac{-1+i}{\sqrt{3}+i}$ in modulus-argument form. 2
- (ii) Hence evaluate $\cos \frac{7\pi}{12}$ in surd form. 2
- (e) Let ω be a non-real cube root of unity.
- (i) Show that $\omega^2 + \omega + 1 = 0$. 2
- (ii) Prove that $b^3 + c^3 = (b+c)(b+c\omega)(b+c\omega^2)$. 2
- (f) Find a polynomial $P(x)$ with real coefficients having $2i$ and $1 - 3i$ as zeroes. 3

Question 2 **16 marks** **(Begin a new booklet)** **Marks**

(a) Find $\int \frac{x}{\sqrt{x+1}} dx$ 2

(b) Solve for x : $\frac{x^2 - 5x}{4 - x} \leq -3$ 3

(c) (i) Sketch the function $f(x) = x^2 - c^2$, where c is a positive constant, clearly indicating its vertex and intercepts. 1

(ii) Hence, without using calculus, draw separate sketches, at least $\frac{1}{3}$ of a page, for each of the following curves. Clearly indicate turning points.

(A) $f(x) = |x^2 - c^2|$ 2

(B) $f(x) = \frac{1}{x^2 - c^2}$ 2

(C) $f(x) = \sqrt{x^2 - c^2}$ 2

(D) $f(x) = (x^2 - c^2)^2$ 2

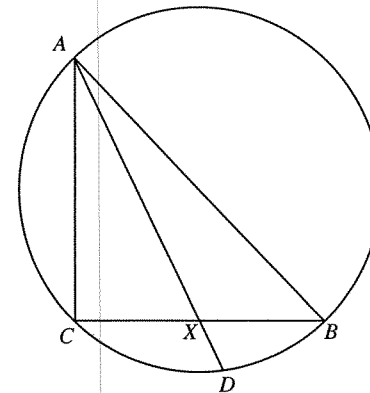
(E) $f(x) = (x^2 - c^2)^3$ 2

Question 3 **16 marks** **(Begin a new booklet)** **Marks**

(a) Consider the function $f(x) = x - \ln(1 + x^2)$. 3

Show that $f'(x) \geq 0$ for all values of x .

(b)



ABC is a triangle inscribed in a circle as shown above. The bisector of $\angle BAC$ meets BC in X and the circle at D .

(i) Prove that $\triangle ABX \sim \triangle ADC$ 2

(ii) Prove that $AB \cdot AC = AD \cdot AX$ 1

(iii) Prove that $AB \cdot AC = AX^2 + BX \cdot XC$ 2

Question 3 continues next page

Question 3 continued

Marks

- (c) A cubic equation has roots
- l, m
- and
- n
- . Given that

$$l + m + n = -3$$

$$l^2 + m^2 + n^2 = 29$$

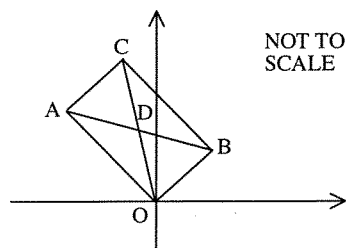
$$lmn = -6$$

- (i) By considering the expansion of
- $(l + m + n)^2$
- , show that the cubic equation is given by: 2

$$x^3 + 3x^2 - 10x + 6 = 0$$

- (ii) Hence find the values of
- l, m
- and
- n
- . 3

- (d) OACB is a rectangle where
- $OA = 2OB$
- . D is the point of intersection of the diagonals. The point B represents the complex number
- z
- .

Find in terms of z , the complex number represented by:

- (i) A 1
- (ii) D 2

Question 4 16 marks (Begin a new booklet)

Marks

- (a) Assuming the result
- $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$
- and using a suitable substitution, solve the equation
- $8x^3 - 6x + 1 = 0$
- . 3

- (b) (i) If
- x
- and
- y
- are real, prove that
- $x^2 + y^2 \geq 2xy$
- . 2

- (ii) Hence show that
- $a^2 + b^2 + c^2 + d^2 \geq 4abcd$
- . 2

- (c)
- $P(2ap, ap^2)$
- is a point on the parabola
- $x^2 = 4ay$
- .
- F
- is the focus of the parabola.
- PQ
- is the perpendicular from
- P
- to the directrix
- d
- , of the parabola. The tangent at
- P
- to the parabola, cuts the axis of the parabola at the point
- R
- .

- (i) Show that the tangent at the point
- P
- to the parabola has equation 2

$$px - y - ap^2 = 0$$

- (ii) Show that
- PR
- and
- QF
- bisect each other. 3

- (iii) Show that
- $PR \perp QF$
- . 2

- (iv) What type of quadrilateral is
- $PQRF$
- ? Give reasons for your answer. 2

Question 5 16 marks (Begin a new booklet)**Marks**(a) Let $y = uv$ be the product of u and v , where u and v are functions of x .

(i) Show that $y'' = uv'' + 2u'v' + u''v$.

2

(ii) Find similar expressions for y''' , y'''' and y''''' .

2

(iii) Hence or otherwise, find and simplify $\frac{d^5}{dx^5} \left((1-x^2)e^{-x} \right)$.

2

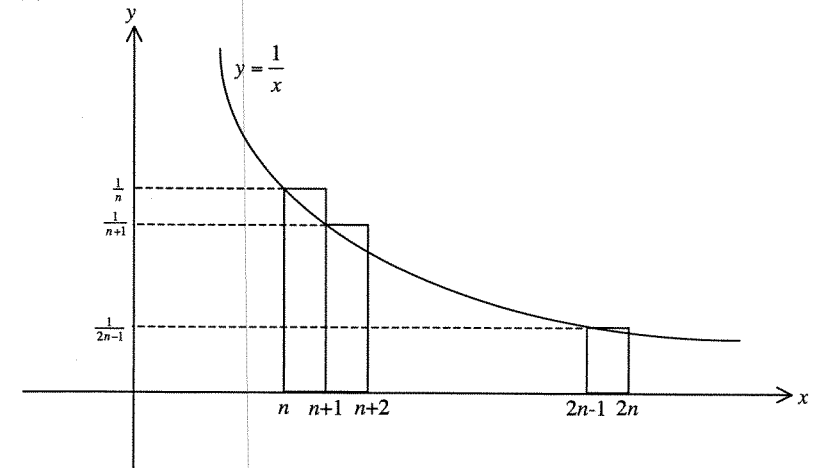
*Question 5 continues next page***Question 5 continued****Marks**

(b) For all integers $n \geq 1$, $t_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1} + \frac{1}{2n}$.

(i) Show that $t_n + \frac{1}{2n} = \frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{2n-1}$

1

(ii)

The diagram above shows the graph of the function $y = \frac{1}{x}$ for $n \leq x \leq 2n$.Using the diagram, show that $t_n + \frac{1}{2n} > \ln 2$.

3

(iii) For all integers $n \geq 1$, let $s_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2n-1} - \frac{1}{2n}$.

Using Mathematical Induction, prove that $s_n = t_n$.

4

(v) Hence find, to three decimal places, the value of:

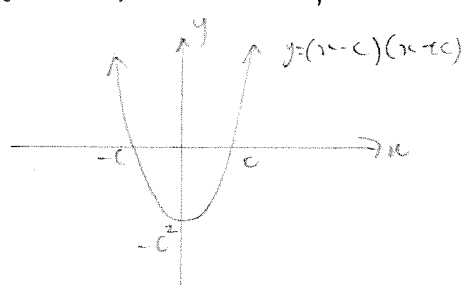
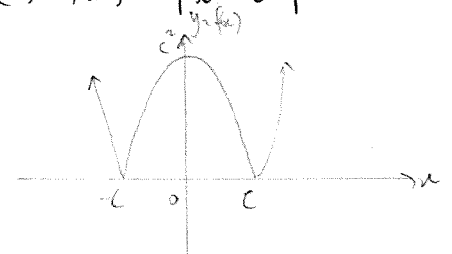
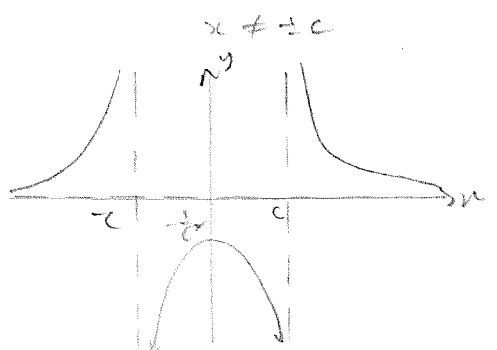
$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{9999} - \frac{1}{10000}$$

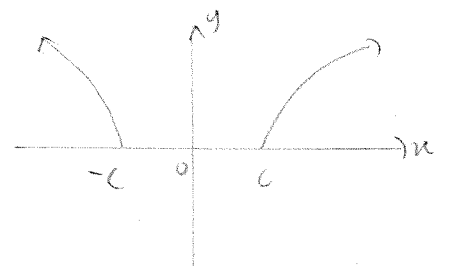
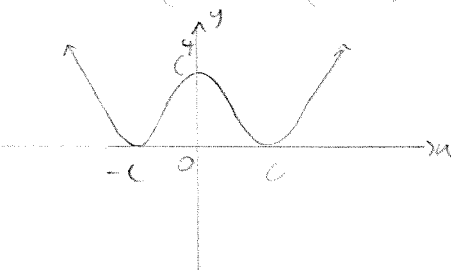
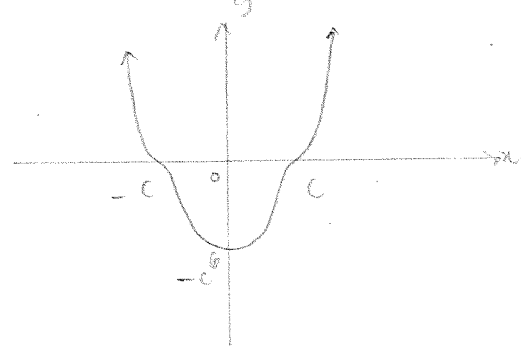
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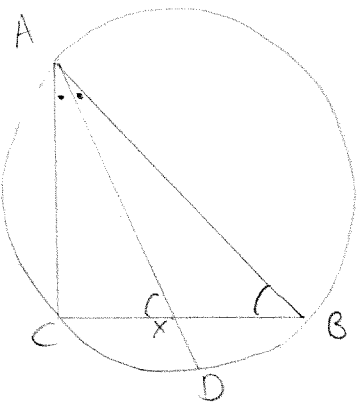
End of Examination

Qn	Solutions	Marks	Comments
	<u>Kambala Extn 2 Half-Yearly Exam 2008</u>		
	<u>Question 1</u>		
(a)	$ 5-2i = \sqrt{5^2+2^2} = \sqrt{29}$	1	
(b)	$z = -3-4i$ $\frac{1}{z} = \frac{1}{-3-4i} \times \frac{-3+4i}{-3+4i}$ $= \frac{-3+4i}{9+16}$ $= -\frac{3}{25} + \frac{4}{25}i$	1	
(c)	$\frac{1+i^5}{1-i}$ $i^4=1$ $i^5=i$ $= \frac{1+i}{1-i} \times \frac{1+i}{1+i}$ $= \frac{1-1+2i}{1+1}$ $= \frac{2i}{2}$ $= i$	1	
(d)	(i) $z = \frac{-1+i}{\sqrt{3}+i}$ $(-1+i) = \sqrt{2} \operatorname{cis} \frac{3\pi}{4}$ $(\sqrt{3}+i) = 2 \operatorname{cis} \frac{\pi}{6}$ $z = \frac{\sqrt{2} \operatorname{cis} \frac{3\pi}{4}}{2 \operatorname{cis} \frac{\pi}{6}}$ $= \frac{\sqrt{2}}{2} \operatorname{cis} \left(\frac{3\pi}{4} - \frac{\pi}{6} \right)$ $= \frac{\sqrt{2}}{2} \operatorname{cis} \left(\frac{7\pi}{12} \right)$ (ii) $\frac{\sqrt{2}}{2} \left(\cos \frac{7\pi}{12} + i \sin \frac{7\pi}{12} \right) = \frac{-1+i}{\sqrt{3}+i}$ $\frac{-1+i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i} = \frac{-\sqrt{3}+i+\sqrt{3}i+1}{3+1} = \frac{1-\sqrt{3}+i(1+\sqrt{3})}{4}$ Equating real parts $\frac{\sqrt{2}}{2} \cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{4}$ $\cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{4} \times \frac{2}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$ $= \frac{\sqrt{2}(1-\sqrt{3})}{4} \text{ or } \frac{\sqrt{2}-\sqrt{6}}{4}$	1	

Qn	Solutions	Marks	Comments
	<u>Question 1 ctd.</u>		
(e)	(i) If w is a non-real cube root of unity, <u>Method I</u> $w^3 = 1$ $w^3 - 1 = 0$ $(w-1)(w^2+w+1) = 0$ $\therefore w = 1$ or $w^2+w+1 = 0$ but w is a non-real cube root of unity $w \neq 1$ so $w^2+w+1 = 0$	1	
	<u>Method II</u> $z^3 = 1$ let $z = 1 \operatorname{cis} \theta$ $\operatorname{cis} 3\theta = \operatorname{cis} 0$ $3\theta = 0 + 2k\pi$ $\theta = \frac{2k\pi}{3}$ when $k=0$ $z_1 = \operatorname{cis} 0 = 1$ $k=1$ $z_2 = \operatorname{cis} \frac{2\pi}{3} = w$ $k=2$ $z_3 = \operatorname{cis} \frac{4\pi}{3} = w^2$ For $z^3 - 1 = 0$ Sum roots = 0 $\therefore 1+w+w^2 = 0$	1	
	<u>Method III</u> $w^3 = 1$ $(w^2)^3 = (w^3)^2 = 1$ $\therefore w^2$ is also a root $1^3 = 1$ and 1 is obviously a root $\therefore w^2, w$ and 1 are cube roots of unity For $z^3 - 1 = 0$ Sum of roots = 0 $\therefore w^2 + w + 1 = 0$	1	
	(ii) $\text{RHS} = (b+c)(b+cw)(b+cw^2)$ $w^3 = 1$ $= (b+c)(b^2 + bcw^2 + bcw + c^2w^3)$ $w^2+w+1=0 \Rightarrow w^2+w=-1$ $= (b+c)(b^2 - bc + c^2)$ $\therefore w^2+w=-1$ $= b^3 + c^3 = \text{LHS}$ Since $x^3+y^3 = (x+y)(x^2-xy+y^2)$ <u>Method II (Similar)</u> LHS $b^3+c^3 = (b+c)(b^2-bc+c^2)$ RHS $= (b+c)(b+cw)(b+cw^2)$ $(b+cw)(b+cw^2) = b^2 + bcw^2 + bcw + c^2w^3$ $= b^2 + bc(w^2+w) + c^2$ $= b^2 - bc + c^2$ $\therefore \text{LHS} = \text{RHS}$	1	

Qn	Solutions	Marks	Comments+Criteria
2 c/d	(c) (i) $f(x) = x^2 - c^2$, $c > 0$ 		
	(ii) (A) $f(x) = x^2 - c^2$ 		
	(B) $f(x) = \frac{1}{x^2 - c^2} = \frac{1}{(x-c)(x+c)}$ $x \neq \pm c$  if $x=0$, $f(x) = \frac{1}{-c^2}$		

Qn	Solutions	Marks	Comments+Criteria
2 c/d	(c) c/d (c) $f(x) = \sqrt{x^2 - c^2}$ 		
	(D) $f(x) = (x^2 - c^2)^2$ $= (x^2 - c^2)(x^2 - c^2)$ $= (x - c)^2 (x + c)^2$ 		
	(E) $f(x) = (x^2 - c^2)^3$ $= (x - c)^3 (x + c)^3$ 		

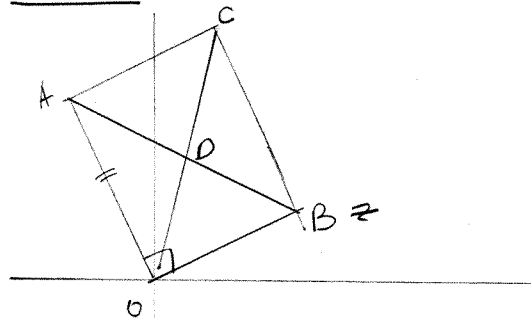
Qn	Solutions	Marks	Comments+Criteria
3	(a) $f(x) = x - \ln(1+x^2)$ $f'(x) = 1 - \frac{2x}{1+x^2}$ $= \frac{1+x^2-2x}{1+x^2}$ $= \frac{x^2-2x+1}{1+x^2}$ $= \frac{(x-1)^2}{1+x^2}$ Since $x^2 > 0$, then $1+x^2 > 0$ $(x-1)^2 > 0 \therefore f'(x) > 0 \forall x$		
	(b)  (i) Prove $\triangle ABX \parallel \triangle ADC$ In $\triangle ABX$ and $\triangle ADC$: AD is bisector of $\angle BAC$ (data) $\therefore \angle CAX = \angle BAX$ $\angle AXC = \angle ABX$ (angles in same arc =) $\therefore \angle ACX = \angle AXB$ (angle sum of $\triangle$) $\therefore \triangle ABX \parallel \triangle ADC$ (equiangular)		

Qn	Solutions	Marks	Comments+Criteria
3	(b) ctd (ii) Prove $AB \cdot AC = AD \cdot AX$ Since $\triangle ABX \parallel \triangle ADC$ from (i), then corresponding sides are in proportion. i.e. $\frac{AB}{AX} = \frac{AD}{AC}$ $\therefore AB \cdot AC = AD \cdot AX$		
	(iii) Prove $AB \cdot AC = AX^2 + BX \cdot XC$ From (ii), $AB \cdot AC = AD \cdot AX$ $= (AX + XD) \cdot AX$ $= AX^2 + XD \cdot AX$ $= AX^2 + BX \cdot XC$ since $AX \cdot XD = BX \cdot XC$ by intercept theorem		

Qn	Solutions	Marks	Comments+Criteria
3	(c) $l+m+n = -3$ = Sum cubic $l^2+m^2+n^2 = 29$ = Sum of squares $lmn = -6$ = Product (i) $(l+m+n)^2$ $= l^2+m^2+n^2 + lm+ln+nl+mn+nl+lm$ $= l^2+m^2+n^2 + 2(lm+mn+nl)$ $(-3)^2 = 29 + 2(lm+mn+nl)$ $9 = 29 + 2(lm+mn+nl)$ $lm+mn+nl = \frac{-20}{2}$ $= -10$ cubic is given by $x^3 - (\text{sum})x^2 + (\text{product of 2})x - \text{product} = 0$ $\therefore x^3 + 3x^2 - 10x + 6 = 0$ as reqd		
	(ii) $x^3 + 3x^2 - 10x + 6 = 0 = P(x)$ $P(1) = 0$. $\therefore x-1$ is a factor. $ \begin{array}{r} x^2 + 4x - 6 \\ x-1 \overline{) x^3 + 3x^2 - 10x + 6} \\ \underline{x^3 - x^2} \\ 4x^2 - 10x \\ \underline{4x^2 - 4x} \\ -6x + 6 \\ \underline{-6x + 6} \\ 0 \end{array} $		

Qn	Solutions	Marks	Comments+Criteria
3	(c) ad $\therefore P(x) = (x-1)(x^2 + 4x - 6)$ $x=1$ a root, $x = \frac{-4 \pm \sqrt{16 - 4 \cdot 1 \cdot (-6)}}{2 \cdot 1}$ $= \frac{-4 \pm \sqrt{16 + 24}}{2}$ $= \frac{-4 \pm \sqrt{40}}{2}$ $= \frac{-4 \pm 2\sqrt{10}}{2}$ $x = -2 \pm \sqrt{10}$ $\therefore l, m, n$ are $1, -2 \pm \sqrt{10}$.		

Q3 (d)



(i) let z be $r(\cos \theta + i \sin \theta)$

$$\vec{OB} = z$$

$$OA = 2OB$$

$$\vec{OA} = 2z i \quad (\text{rotation anticlockwise by } \frac{\pi}{2})$$

$$\therefore A \text{ is } 2iz$$

$$(ii) \vec{OC} = \vec{OB} + \vec{BC}$$

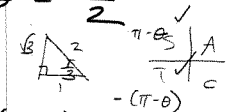
$$= z + 2iz$$

$$= (1+2i)z$$

$$\therefore \vec{OB} = \frac{1}{2}(1+2i)z$$

$$\therefore D \text{ is } \frac{1}{2}(1+2i)z$$

$$\text{or } \left(\frac{1}{2} + i\right)z$$

Qn	Solutions	Marks	Comments
	<u>Question 4</u> (a) $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ if $8x^3 - 6x + 1 = 0$ $8x^3 - 6x = -1$ $4x^3 - 3x = -\frac{1}{2}$ Looking for 3 solutions (unique) Let $x = \cos \theta$ $4\cos^3 \theta - 3\cos \theta = -\frac{1}{2}$ $\cos 3\theta = -\frac{1}{2}$  $3\theta = \pi - \frac{\pi}{3}, -(\pi - \frac{\pi}{3})$ $= \frac{2\pi}{3} + 2k\pi, -\frac{2\pi}{3} + 2k\pi$ $3\theta = \frac{6k\pi \pm 2\pi}{3} = \frac{2\pi(3k \pm 1)}{3}$ $\theta = \frac{2\pi(3k \pm 1)}{9}$ when $k=0$ $\theta = \pm \frac{2\pi}{9} \therefore x = \cos \frac{2\pi}{9}$ $x = \cos(-\frac{2\pi}{9}) = \cos \frac{2\pi}{9}$ when $k=1$ $\theta = \frac{2\pi \pm 4}{9} = \frac{8\pi}{9} \therefore x = \cos \frac{8\pi}{9}$ $\theta = \frac{2\pi \times 2}{9} - \frac{4\pi}{9} \therefore x = \cos \frac{4\pi}{9}$ when $k=2$ $\theta = \frac{2\pi \times 7}{9} - \frac{4\pi}{9} = \frac{10\pi}{9} - \frac{4\pi}{9} = \frac{6\pi}{9} = \frac{2\pi}{3}$ $x = \cos \frac{2\pi}{3} = -\frac{1}{2}$ $\theta = \frac{2\pi \times 5}{9} = \frac{10\pi}{9} - \frac{8\pi}{9} = \frac{2\pi}{9}$ $x = \cos \frac{2\pi}{9}$ $\therefore x = \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}$	1	

Qn	Solutions	Marks	Comments+Criteria
4	(c) ad (ii) Show PR $\perp$ OF $M_{PR} = \frac{ap^2 + ap^2}{2ap - 0}$ $= \frac{2ap^2}{2ap}$ $= p \quad (\text{already shown in (i)!})$ $M_{OF} = \frac{-a - a}{2ap - 0}$ $= \frac{-2a}{2ap}$ $= -\frac{1}{p}$ $M_{PR} \times M_{OF} = p \times -\frac{1}{p}$ $= -1$ $\therefore PR \perp OF$		
	(iv) PQRF? Diagonals bisect @ right angles, $\therefore$ PQRF is a rhombus. FR $\parallel$ PQ since PQ is perp. to directrix, a horizontal line.		

Qn	Solutions	Marks	Comments+Criteria
5	(a) $y = uv$ (i) $y' = u'v + v'u$ $y'' = u''v + v'u' + v''u + u'v'$ $= uv'' + 2u'v' + u''v$		
	(ii) find y''', y'''' $y''' = \underbrace{u'v''} + v'''u + \underbrace{2u''v'} + v'' \cdot 2u' + \underbrace{u''v} + v'u''$ $= uv''' + 3u'v'' + 3u''v' + u'''v$ $y'''' = \underbrace{u'v'''} + v''''u + \underbrace{3u''v''} + v'''' \cdot 3u' + \underbrace{3u''v'} + v'' \cdot 3u'' + \underbrace{u''''v} + v'u''''$ $= uv'''' + 6u''v'' + 4u'v''' + 4u'''v' + u''''v$ $y'''' = u'v'''' + v''''u + 6u''v'' + v'''' \cdot 6u' + 4u''v''' + v'''' \cdot 4u'' + u''''v + v'u''''$ $= uv'''' + 5u'v'''' + 10u''v'' + 10u''v'' + 5u''''v' + u''''v$		$\begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & & 2 & & \\ & & & 3 & \\ & & & & 4 \end{pmatrix}$ $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 9 \end{pmatrix}$
	(ii) $\frac{d^2}{dx^2} \left(\frac{(1-x^2)e^{-x}}{u} \right)$ $u = 1-x^2$ $\frac{du}{dx} = u' = -2x$ $u'' = -2$ $u^3 = u^4 = u^5 = 0$ $y^5 = uv^5 + 5u'v^4 + 10u''v^3 + 10u'''v^2 + 5u''''v' + u''''v$ $\therefore \frac{d^5}{dx^5} \left((1-x^2)e^{-x} \right)$ $v = e^{-x}$ $v'' = e^{-x}$ $v^4 = e^{-x}$ $v' = -e^{-x}$ $v''' = -e^{-x}$ $v^5 = -e^{-x}$ $= (1-x^2)(-e^{-x}) + 5(-2x)e^{-x} + 10(-2)(-e^{-x})$ $+ 10 \cdot 0 \dots + 5 \times 0 \dots + 0$ $= -e^{-x}(1-x^2 + 10x - 20)$ $= -e^{-x}(-x^2 + 10x - 19) = e^{-x}(x^2 - 10x + 19)$		

Qn	Solutions	Marks	Comments+Criteria
5	(b) $t_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1} + \frac{1}{2n}$ (i) $t_n + \frac{1}{2n} = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1} + \frac{1}{2n} + \frac{1}{2n}$ $= \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1} + \frac{2}{2n}$ $= \frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{2n-1}$ as required (ii) $\int_n^{2n} \frac{1}{x} dx \approx n \times \frac{1}{n} + n \times \frac{1}{n+1} + n \times \frac{1}{n+2} + \dots + n \times \frac{1}{2n-1}$ $= n \left(\frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{2n-1} \right) = \left(t_n + \frac{1}{2n} \right)$ also $\int_n^{2n} \frac{1}{x} dx = [\ln x]_n^{2n}$ $= \ln 2n - \ln n$ $= \ln 2 + \ln n - \ln n$ $= \ln 2$ From diagram, area of rectangle is an approximation yields an area greater than exact area. $\therefore \left(\frac{1}{n} + \frac{1}{n+1} + \dots + \frac{1}{2n-1} \right) > \ln 2$ i.e. $\left(t_n + \frac{1}{2n} \right) > \ln 2$		

Qn	Solutions	Marks	Comments+Criteria
5	(iii) $S_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2n-1} - \frac{1}{2n}$ Prove $S_n = t_n$. $t_n = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1} + \frac{1}{2n}$ RTP: $1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2n-1} - \frac{1}{2n} = \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{2n-1} + \frac{1}{2n} \text{ for } n \geq 1.$ test $n=1$: LHS = $1 - \frac{1}{2} = \frac{1}{2}$ RHS = $\frac{1}{1+1} = \frac{1}{2} = \text{LHS}$ $\therefore$ true for $n=1$. Assume true for $n=k$ i.e. $S_k = t_k$. i.e. $1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2k-1} - \frac{1}{2k} = \frac{1}{k+1} + \frac{1}{k+2} + \dots + \frac{1}{2k-1} + \frac{1}{2k}$ Prove true for $n=k+1$ i.e. $S_{k+1} = t_{k+1}$. i.e. RTP: $1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2(k+1)-1} - \frac{1}{2(k+1)} = \frac{1}{(k+1)+1} + \frac{1}{(k+1)+2} + \dots + \frac{1}{2(k+1)}$ i.e. $1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2k+1} - \frac{1}{2k+2} = \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k+2}$ LHS = $1 - \frac{1}{2} + \frac{1}{3} - \dots + \frac{1}{2k+1} - \frac{1}{2k+2}$ $= \underbrace{1 - \frac{1}{2} + \frac{1}{3} - \dots - \frac{1}{2k}}_{S_k} + \frac{1}{2k+1} - \frac{1}{2k+2}$ $= \frac{1}{k+1} + \frac{1}{k+2} + \dots + \frac{1}{2k-1} + \frac{1}{2k} + \frac{1}{2k+1} - \frac{1}{2k+2}$ $= \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k+1} + \frac{2k+2 - (k+1)}{(2k+2)(k+1)}$ $= \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k+1} + \frac{k+1}{(2k+2)(k+1)}$ $= \frac{1}{k+2} + \frac{1}{k+3} + \dots + \frac{1}{2k+1} + \frac{1}{2k+2}$ = RHS $\therefore$ true for $n=1, \dots$		

Qn	Solutions	Marks	Comments+Criteria
5 (iv)	$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{9999} - \frac{1}{10000}$ $S_n = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2n-1} - \frac{1}{2n}$ So $2n = 10000$ $n = 5000$ $t_n = S_n \therefore t_{5000} = S_{5000}$ $t_n + \frac{1}{2n} > \ln 2$ $t_{5000} + \frac{1}{10000} > \ln 2$ $t_{5000} > \ln 2 - \frac{1}{10000}$ $> 0.693047\dots$ $\therefore t_{5000} \approx 0.693$		